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BUSINESS PLAN 2.0

DISRUPTIVE LAUNCH

ColdProof

***A Cold-Chain Temperature-Monitoring and Audit-Compliance Platform
for the United Arab Emirates***

Master of Global Business (MGB)

Course: Entrepreneurship for Disruptive Launch

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Executive Summary

ColdProof (working name) is a subscription-based software solution which allows temperature and humidity monitoring of cold rooms and refrigerated vehicles in real time and alerts the operator when a temperature excursion will cause products to go bad, and documents tamper-proof and audit-proof documentation for food-safety inspectors and pharmaceutical buyers. It's not about refrigeration, it's about software: The venture sells low-cost sensors by trusted vendors to its platform and the defensible asset is the compliance history.

This is because the United Arab Emirates (UAE) are the launching country for 3 reasons. It is dependent on imports for a significant proportion of its food supply, and consequently has a large temperature controlled supply chain; food establishments are required to have a mandated food-safety regime based on Hazard Analysis and Critical Control Point (HACCP) principles with legally required temperatures and record keeping (Dubai Municipality, 2013); and government and private operators are heavily investing in cold storage infrastructure (Mordor Intelligence, 2026). Version one provides three capabilities: live monitoring, excursion alerts and audit records, all charged on a per monitored point, per month basis, plus pharmaceutical lanes available in an add-on per shipment.

A mid-size operator, having 200 monitored points, will enjoy about AED 96,000 in annual recurring revenue. After passing sensor hardware through, ten of these accounts will generate roughly AED 4.8-6.0 million in recurring revenue within 5-7 years with a gross margin of 70-80%. The estimated construction cost for v1 is AED 0.8–1.5 million. The dependency on third-party hardware, competition in the pharmaceutical sector and interoperability challenges documented in the cold-chain literature (Ahmad et al. 2024) are the main risks, and mitigations have been provided for each respective risk in Section 11. The financial information contained in this report comes from the venture's own projections, and is clearly marked throughout from other evidence provided.

1. The Start-Up Idea

The range of temperatures in cold chains is very narrow. The frozen product is stored below -18°C , the chilled product is stored between $+2^{\circ}\text{C}$ and $+8^{\circ}\text{C}$, and many pharmaceuticals are stored within even narrower limits. A brief aberration – a door left open, a compressor malfunction or a delivery truck parked in the summer heat – can mar the product as well as the audit trail. The operator suffers two losses here – the lost load and not being able to establish that the remaining cargo was not lost.

The magnitude of the issue at hand is very clear. According to the Food and Agriculture Organization and the United Nations Environment Programme, the lack of refrigeration contributes to the loss of approximately 526 million tonnes of food annually, representing approximately 12% of world's total food production, and that about 14% of food is lost before being consumed (FAO & UNEP, 2022). According to World Bank estimates (2021), in developing economies, at least 25 per cent of food production is being lost due to the lack of a good cold chain. The World Health Organization (WHO, 2015) states that majority of vaccines require $+2^{\circ}\text{C}$ and $+8^{\circ}\text{C}$ with continuous monitoring of the temperature throughout the entire supply chain, while Jiang et al. (2024) state that sub-optimal temperatures in the transport and storage process are the reason why there is an increase in vaccine related adverse events in the developing world.

The proof of that record being fragmented is just as evident. In a systematic review of 97 peer-reviewed papers, Muller et al. (2026) describe the conventional cold-chain monitoring as “reactive” and “fragmented”, relying on separate technologies instead of an integrated real-time system. In reality operators use disconnected data loggers, manual temperature checks, and multiple separate spreadsheets for each facility. If a food inspector or a pharmaceutical buyer requires an operator to prove that a particular shipment did not exceed its given range, it takes time to put together evidence and often it is not always complete.

ColdProof directly tackles this need. It is a subscription-based solution that combines temperature and humidity data from all the cold rooms and refrigerated vehicles on to one dashboard, notifying the responsible manager before any drift affects compliance, and generates the compliance report that regulators and buyers demand. It is not a sensor manufacturer nor does it have its own refrigeration system. It controls the software layer and most importantly, the compliance record on which the rest of the supply chain is dependent.

2. How It Works

Version one is designed to do just three things, which is why the product can get to a live operating site fast.

- **Live monitoring.** LoRaWAN or Bluetooth gateways provided by known manufacturers, deliver temperature and humidity data from off-the-shelf sensors installed in every room or vehicle to one dashboard. An operator can now view all zones at all sites in one image, instead of walking the floor with a thermometer.
- **Excursion alerts.** Before exceeding the limit, the platform alerts you of a drift towards the limit and alerts the duty manager to the risk via SMS or in-app notification. Reliance on few trained staff is reduced, and errors and losses of manual checks are reduced, due to continuous automatic alerting (Jiang et al., 2024).
- **Audit records.** Each reading is recorded and kept tamper evident and can be exported as a compliance report per room, per shipment, with a timestamp as a food

inspector or pharmaceutical buyer would expect. This function provides the customer retention for the venture. Other functions such as predictive maintenance for compressors, energy optimisation and integration with the operator's warehouse-management system will be left to later releases. The review evidence shows that the automatic control of data as well as models using sensors is rarely realized, and is indicated as a future research direction instead of a proven skill (Müller et al., 2026). The platform is set up to monitor the temperature bands as shown in Table 1.

Product category	Typical temperature band	Primary failure consequence
Frozen food	Below -18°C	Thaw, refreeze, texture and microbial spoilage
Chilled food (high-risk/perishable)	At or below $+5^{\circ}\text{C}$ (UAE food law)	Bacterial growth, food-poisoning risk
Pharmaceuticals / vaccines	$+2^{\circ}\text{C}$ to $+8^{\circ}\text{C}$ (most vaccines)	Loss of potency, consignment rejection
Ultra-low-temperature biologics	As low as -70°C	Total product loss

Table 1. Temperature bands monitored by the platform. Food thresholds follow UAE food law (Dubai Municipality, 2013); vaccine range follows WHO (2015).

3. Value Proposition

The platform for a food operator is a remedy to prevent food losses due to spoilage and a source of the required documentation for a municipal food-safety audit. UAE food law stipulates that high-risk foods and perishable foods must be kept at or below $+5^{\circ}\text{C}$, a documented programme of food-safety (HACCP) is required and that records of temperatures and corrective action must be kept and made available to a food inspector when requested (Dubai Municipality, 2013). The platform automates just this record. A platform can safeguard high-value consignment and provides the continuous-monitoring data that Good Distribution Practice and WHO guidance call for (WHO, 2015) for a pharmaceutical distributor or third-party logistics service provider. The return on investment case is a short one as a rejected biologics consignment can be a big multiple of the annual subscription. The principle commercial argument is that a single instance of a potential spoilage event covers months or, in the pharmaceutical example, years of subscription. Even just one damaged pallet of seafood or a rejected pharmaceutical shipment is worth tens of thousands of dirhams, and one yearly fee is less than that. This is a guess as to how much customers will pay for this service built in the first reference deployment, and is not a fact that can be substantiated or verified from outside the venture.

4. Market Size and Potential

4.1 Global and regional context

The climate of the entire country or region. Overall national or regional climate conditions. The estimates presented here are indicative range estimates and are not accurate numbers due to the different scope and methodologies used for calculating the cold-chain monitoring

market. According to one estimate, the worldwide cold-chain monitoring market is projected to reach USD 15.04 billion in 2030 with a compound annual growth rate (CAGR) of 12.6%, and regulatory compliance is one of the most prominent market drivers, with software being the fastest growing offering (MarketsandMarkets, 2025). According to a separate analysis, the software segment is expected to grow at a higher rate as compared to the market at 20.9% CAGR, while the frozen band is the major temperature segment (Grand View Research, 2024). The range in published estimates is recorded in Appendix B and the reader should evaluate the uncertainty. The significance of these numbers to ColdProof is that the software layer is the fastest growing and highest margin portion of the market—this is where the venture sits.

4.2 The UAE market

The UAE cold-chain logistics market size was valued at around USD 1.4 billion in 2024 and is expected to rise to USD 2.9 billion by 2033 at 8.20% CAGR (IMARC Group, 2025). According to a second source, the 2024 figure is between USD 2.5 billion and 2032, increasing by 7.3% (Verified Market Research, 2025), which again attributes to the sector's boundary. Key drivers of demand in the UAE include a high reliance on food imports, expanding online grocery delivery services, pharmaceutical and vaccine distribution, and continued investment in cold-storage facilities by both governments and private operators, such as a reported AED 621 million cold-storage programme by KEZAD Group (Mordor Intelligence, 2026).

Market opportunity: TAM → SAM → SOM (UAE beachhead)

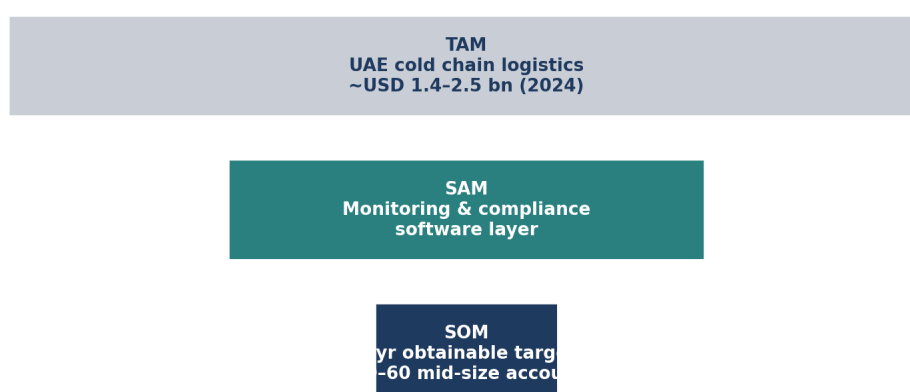


Figure 1. Market opportunity framed as TAM – SAM – SOM for the UAE beachhead. TAM figure from IMARC Group (2025) and Verified Market Research (2025).

4.3 Segmentation, pricing, and product use

There are two types of buyers in the market and the entry point of the venture is what makes it go either one way or the other. The food side, which includes cold-storage operators, food importers, distributors of dairy, seafood, meat and produce, is the larger market, with a quicker buying decision, and with lower stakes per shipment; it's the recommended beachhead, as pharmaceutical-grade competitors under-serve it. The pharmaceutical side, which includes distributors and third-party logistics companies that transport medicines, vaccines and biologics, is smaller in number of customers but higher in value of each shipment due to the high cost of a lost shipment and a failed Good Distribution Practice audit that could jeopardise a licence. The segmentation is given in Table 2.

Dimension	Food side (beachhead)	Pharmaceutical side (phase two)
Buyers	Cold-storage operators, importers, distributors	Pharma distributors, 3PLs, hospital supply
Core fear	Spoilage loss; failing a municipal food-safety audit	Consignment rejection; failing a GDP audit
Sales cycle	Faster	Longer, more technical
Willingness to pay	Moderate	High
Competition	Under-served by pharma-grade vendors	Contested by global players

Table 2. Market segmentation and entry sequence.

The pricing is based on a "per monitored point per month" basis, with a monitored point being either a room, chamber or vehicle. The indicative range is AED 30 – 80 per point per month, with a minimum fee of AED 30 at the facility, and an additional fee of AED 50 – 200 per shipment for pharmaceutical lanes where documentary proof per shipment becomes the most critical. For a mid-size operator with 150-250 monitored points, the cost would be around AED 6,000 to 15,000 per month. This is a structure that guarantees the venture stays within recurring software income, and permits for the account to develop as the owner incorporates rooms and vehicles.

5. Other Market Segments

There are several contiguous segments that have the same underlying need (continuous temperature evidence) and can be considered expansion sites after the food and pharmaceutical beachheads are secured.

They are of lesser importance for Version one and they are mentioned here for the sake of completion. Floriculture, which imports cut flowers and plants, is another segment which needs more stringent control of temperature, and UAE cold-chain operators are already catering to this segment. Specialty chemicals and high-end electronics that require stability and/or warranty storage.

Blood products and clinical-trial biologics, in which the chain of custody requirements are even more stringent than for regular pharmaceuticals. Multi-site chilled and frozen storage systems will be subject to the same municipal food-safety audits as quick-service restaurant and hotel chains. E-grocery dark stores and micro-fulfilment centres, a growing UAE category that merges chilled, frozen and ambient zones (Mordor Intelligence, 2026).

6. Key Stakeholders, Industry and Competitor Analysis

6.1 Stakeholders

The operator's owner or general manager is the buying stakeholder and authorises the subscription while the duty or quality manager gets alerts and pulls reports is the daily user. The standards the record will meet are determined by the regulators, which in this

case are the Food Control function of Dubai Municipality for food, and the pharmaceutical regulator for medicines (Dubai Municipality, 2013; WHO, 2015). Sensor manufacturers are supply partners rather than competitors. Downstream stakeholders such as pharmaceutical buyers and third-party logistics (TPL) providers are also stakeholders whose quality demands pull the record through. Insurers for perishable cargo – they are a potential distribution and validation partner, as there is a reduced risk of claims, when verifiable temperature records are involved.

6.2 Competitor landscape

The market is divided as many global specialist vendors are present along with the in-house tools developed by large logistics service providers (MarketsandMarkets, 2025; Grand View Research, 2024). Specialists like Sensitech (Carrier), ELPRO, Controlant, ORBCOMM, Berlinger, Monnit and DeltaTrak are well established in the pharmaceutical monitoring and global shipping lanes market. The interpretation the venture makes is that these players focus on high value pharmaceutical/export applications and neglect the mid-market in UAE which is quicker to make a purchase decision and where the pricing of incumbents is not as competitive; it is an interpretation, not a fact. The landscape is summarised in Table 3.

Competitor type	Strength	Gap ColdProof targets
Global pharma-grade vendors	Deep pharma compliance, global lanes	Cost and fit for UAE food mid-market
Logistics providers' in-house tools	Tied to their own operations	Not sold as an open, cross-operator record
Standalone data-logger suppliers	Cheap hardware	No consolidated dashboard or audit report
ColdProof (proposed)	UAE-configured compliance record, per-point SaaS	Building food-side reference base first

Table 3. Competitor landscape and targeted gap.

7. Distribution Channels

The early motion is direct B2B sales into one live facility with continuing expansion within that operator as rooms and vehicles are added (Often called a land and expand model which works well with per point pricing). It is more valuable than any pitch document to get a warm introduction into one operating site than to have it in any pitch document, as the entire early task is getting sensors into an operating facility and proving that the alerts work. In addition to direct sales, four channels cover the reach: partnerships with sensor and gateway suppliers who can combine the software with their hardware; referral relationships with food-safety consultants and auditors who advise operators on specifically the compliance gap the platform closes; access to pharmaceutical accounts via third-party logistics providers once food-side references have been established; and partnerships with cargo insurers for whom verifiable records reduce claims exposure. The channels are arranged in a sequence for the food side direct sale to prove the product before the channels of the pharmaceutical and partner products are opened.

8. Funding and Risk Analysis

8.1 Financing requirement and options

This paper discusses the financing requirement and options for 8.1. Version one is estimated to cost AED 0.8–1.5 million (Section 10). The financing issue is how to fund, for about 18 months, the engineering work and early sales of the venture—without handing over too much equity before the food side is proven to be viable. Table 4 presents a comparison of the realistic sources and the cost, control, speed, and fit.

Financing source	Cost / dilution	Speed & fit for this stage
Founder equity / bootstrapping	No dilution; limited by personal capital	Immediate; suitable for first prototype only
Angel investors	Moderate dilution	Good fit for a pre-seed round at this stage
Government-linked incubators / accelerators / grants	Low or no dilution (non-dilutive support)	Strong fit in the UAE; slower to secure
Venture capital	Higher dilution; board influence	Better deferred until food-side traction exists
Revenue-based financing	No dilution; repaid from revenue	Only viable once recurring revenue is live
Strategic partner investment (sensor supplier / 3PL)	Dilution plus channel commitment	Useful later for distribution, not at seed
Bank debt	No dilution; requires security / cash flow	Poor fit pre-revenue for an asset-light firm

Table 4. Comparison of financing sources for the pre-seed stage.

The model is to conduct a small pre-seed investment by angel investors and then receive non-dilutive support from a UAE government-backed accelerator or grant programme in exchange for paying reference accounts to validate the food-side model, before securing a VC investment. This maintains founder control during the most uncertain time and also represents the lowest to medium level of capital intensity of the build. Revenue-based financing can become a viable, non-dilutive approach to scaling sales, as recurring revenue becomes available.

8.2 Opportunities and challenges

The main opportunities are structural. Compliance needs – compliance rules stimulate adoption of continuous monitoring of food and pharmaceutical supply chains (MarketsandMarkets, 2025) – UAE food law already requires continuous monitoring records (Dubai Municipality, 2013). The software segment makes up for the highest gross margin and also the fastest-growing segment of the market (2024), according to Grand View Research. The more monitoring information that has been gathered the more effective the alerting becomes, and a new party is unlikely to be able to duplicate this. These issues include a long business-to-business sales cycle and the need to get a "reference site" to

unlock the sale, reliance on third-party hardware, and interoperability and cost issues discussed in Section 11.

9. Key Metrics for Revenue Generation

The per-point subscribed to the core revenue stream and the per-shipment add-on for the pharmaceutical lanes. The metrics the business is managed against are the number of monitored points, monthly and annual recurring revenue (MRR and ARR), average revenue per account (ARPA), net revenue retention, gross margin, customer-acquisition-cost payback, and churn. The unit economics for a representative mid-size account are outlined in Table 5.

Metric	Assumption / value	Basis
Monitored points per account	200	Mid-size operator (range 150–250)
Price per point per month	AED 40	Mid-point of AED 30–80 range
Monthly recurring revenue / account	AED 8,000	$200 \times \text{AED } 40$
Annual recurring revenue / account	AED 96,000	$\text{AED } 8,000 \times 12$
Pharma add-on	AED 50–200 / shipment	Applied to pharma lanes only
Gross margin	70–80%	Cloud + support; hardware passed through

Table 5. Unit economics for a representative mid-size account (venture projection).

The revenue path is provided by scaling this account economics. Each of the 50 food-side accounts in the portfolio is worth \$96,000 per year, before factoring in any pharmaceutical add-ons or multi-site expansion, which equates to around \$4.8 million in annual recurring revenue. Figure 2 is a five-year projection with an increase in the number of accounts from 6 to 60 and an increasingly higher revenue per account in the form of pharmaceutical add-ons. These numbers are the projections of the venture based on the stated assumptions, rather than numbers based on external data to create forecasts.

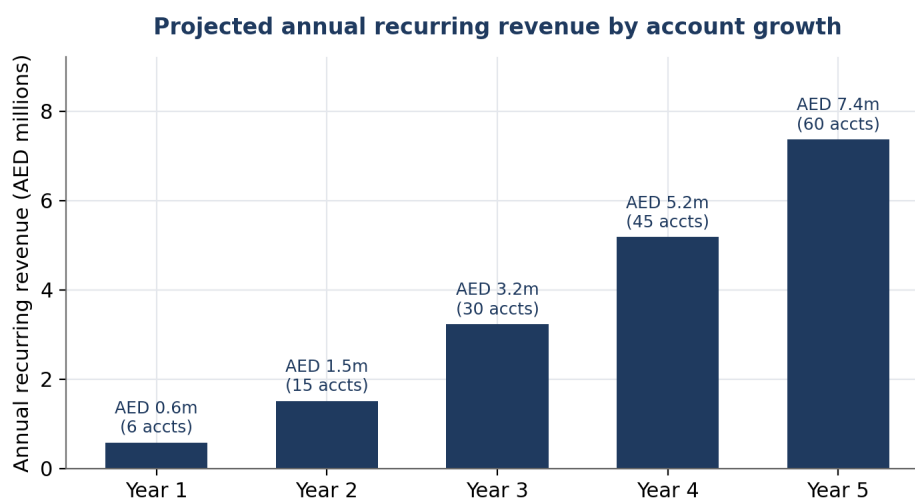


Figure 2. Projected annual recurring revenue by account growth (venture model; assumptions in Appendix A).

10. Cost Structure (Investments)

Version one is supposed to cost AED 0.8 – 1.5 million, and is more software oriented than hardware. The decision against putting in sensors (i.e. established LoRaWAN or Bluetooth gateways) prevents a massive capital investment, and ensures speed as no benefit can be gained from investing in sensors. The indicative allocation is provided in Table 6.

Investment area	Indicative share	Notes
Software engineering	45–55%	Dashboard, alerting engine, compliance-report generator
Cloud infrastructure	8–12%	Scales sub-linearly with customers
Sensor hardware (pass-through)	10–15%	Amortised or billed to customer
Sales and onboarding	12–18%	Reference-site acquisition and deployment
Support and success	6–10%	Retention of early accounts
Compliance and certification	4–8%	Alignment to food and GDP audit templates

Table 6. Indicative version-one investment allocation (venture estimate, AED 0.8–1.5 million total).

The gross margin is estimated at 70-80% after passing through or amortising the sensor hardware. The asset light model is the main appeal of the software layer over the cold storage business: The software business doesn't have to make the significant capital investment in building warehouses, but it does sell to the operators who have already made the investment.

11. Risks and Challenges

The following Table 7 outlines the main risks and mitigations. The two are related to the cold-chain literature, as high implementation cost is cited as a barrier in a review of evidence (MarketsandMarkets, 2025) when small and medium operators are considered, and interoperability between devices and enterprise systems, is cited as a key barrier in a formal barrier analysis of IoT in the cold supply chain (Ahmad et al., 2024).

Risk	Description	Mitigation
Hardware dependence	Reliance on third-party sensors and gateways	Qualify multiple suppliers; pass through cost; own the software layer

Risk	Description	Mitigation
Interoperability / data silos	Difficulty integrating with devices, WMS and ERP (Ahmad et al., 2024)	Open APIs and standard protocols (LoRaWAN, BLE); staged integration
High cost deters SMEs	Advanced monitoring seen as costly (MarketsandMarkets, 2025)	Subscription with facility minimum; one-avoided-loss ROI framing
Cybersecurity	Connected systems raise breach and tampering risk	Encryption, access control, tamper-evident logging
Crowded pharma segment	Global vendors contest pharmaceutical accounts	Lead with the under-served food side; enter pharma with references
Regulatory fragmentation	Requirements differ by emirate and across the GCC	Configurable report templates mapped to each authority
Adoption / skilled-labour reliance	Sector faces a skilled-workforce shortage (Mordor Intelligence, 2026)	Automation reduces manual checks and dependence on scarce staff (Jiang et al., 2024)
Long sales cycle	B2B decision needs a reference site	Land-and-expand; prove alerts in one live facility first

Table 7. Principal risks and mitigations.

12. Conclusions

The evidence brought together here is sufficient to suggest the viability of the venture. It helps sell recurring high-margin software into a UAE cold-chain market that grows at about 7–8%, year-on-year (IMARC Group, 2025) without the hefty overhead of owning refrigeration. The demand is not speculative, but based on documented problems, such as food losses due to inadequate refrigeration (FAO & UNEP, 2022) and strict and legally required temperature and record-keeping requirements in UAE (Dubai Municipality, 2013; WHO, 2015). The cost of compliance is costs incurred by an operator that may be needed if the system is taken away ahead of an inspection, as this audit evidence will not be available without incurring these costs. There are two real questions here: Willingness to pay in the food segment (which is taken as a given in the value proposition, but it hasn't been proved); and competition in the pharmaceutical segment, where there are established global suppliers. The recommended sequence is to prove the food-side model first, before entering the contested pharmaceutical segment, both of them.

13. Recommendations

- Start with the foods. Win a live reference facility and demonstrate the alerts and audit report before pitching pharmaceutical accounts.
- Fund conservatively. Raise a small pre-seed with angel investors plus other non-dilutive accelerator/grant funding and await VC funding when there is some degree of traction in the food side.
- Ship three functions only. Create live monitoring, excursion alerts and audit records; do not consider implementing predictive maintenance or WMS integration until the core is adopted.
- Partner rather than build hardware. Gain access to one or two sensor suppliers rather than manufacturing, save capital and time. Design from the outset for the audit. Switching costs are built in early by reporting to Dubai Municipality food-safety and Good Distribution Practice templates in the

map. • Instrument the business. The unit economics in Section 9 will be validated by tracking ARR, net revenue retention, and customer-acquisition-cost payback from the first account.

14. Strategic Frameworks

This section applies four entrepreneurship frameworks to ColdProof, consolidating the analysis developed in Sections 1–13. Where a cell states a judgement rather than a sourced fact, it reflects the venture's own view and is to be tested in early deployments.

14.1 Problem Statement Canvas

Element	ColdProof
Core problem	A short temperature excursion in a cold room or reefer spoils product and breaks the audit trail at the same time, so the operator loses the cargo and cannot prove the rest of the load stayed in range.
Target user / customer	UAE cold-storage operators, food importers and distributors (beachhead); pharmaceutical distributors and 3PLs (phase two). The daily user is the duty or quality manager.
Current situation	Disconnected data loggers, manual clipboard checks, and separate spreadsheets per facility; assembling proof for an inspector is slow and often incomplete (Müller et al., 2026).
Impact of the problem	Spoiled cargo worth tens of thousands of dirhams per event; failed municipal or GDP audits risking fines or licence loss; large food loss from inadequate refrigeration at sector level (FAO & UNEP, 2022; World Bank, 2021).
Desired outcome	One dashboard showing every zone in real time, an alert before a breach, and an audit-ready compliance record produced on demand.
Why this matters now	UAE food-import dependence, legally mandated HACCP-based temperature records (Dubai Municipality, 2013), a cold-chain market growing 7–8% a year (IMARC Group, 2025), and a skilled-labour shortage that makes manual compliance harder each year (Mordor Intelligence, 2026).
Summary problem statement	UAE cold-chain operators need to prove, continuously and on demand, that temperature-sensitive product stayed within range — because a single excursion costs them both the cargo and the audit evidence, and current manual methods cannot deliver that proof reliably.

Table 8. Problem Statement Canvas for ColdProof.

14.2 Lean Canvas

The nine blocks below follow the standard Lean Canvas, with the common mistakes flagged in the course template avoided — the problem block describes problems rather than the solution, the segment names a specific early adopter, and the unfair advantage is a structural asset rather than “our team.”

Block	ColdProof
Problem	(1) Excursions spoil product before anyone reacts; (2) proof of compliance is fragmented across loggers, clipboards and spreadsheets; (3) skilled compliance staff are scarce.
Customer segments	Early adopter: mid-size UAE cold-storage operators and food distributors running 150–250 monitored points. Later: pharmaceutical 3PLs.

Block	ColdProof
Unique value proposition	The trusted, audit-ready record that proves refrigeration worked — and the alert that fires before it fails.
Solution (MVP)	(1) Live multi-site dashboard; (2) pre-breach excursion alerts by SMS and app; (3) tamper-evident, exportable compliance reports.
Channels	Primary: direct B2B land-and-expand from one live reference facility. Secondary: sensor-supplier bundles, auditor referrals, 3PL access.
Revenue streams	Per-monitored-point subscription (AED 30–80 per point per month) plus a per-shipment pharmaceutical add-on (AED 50–200).
Cost structure	Software engineering (dashboard, alerting engine, report generator), cloud infrastructure, sensor pass-through, sales and onboarding (CAC), and support.
Key metric	Net revenue retention on monitored points — the single number showing whether accounts expand as operators add rooms and vehicles.
Unfair advantage	Compliance lock-in: an operator will not remove the system that produces its audit evidence before an inspection. Deepened by accumulated excursion data and report templates mapped to each regulator.

Table 9. Lean Canvas for ColdProof.

14.3 Stakeholder Map

Influence and interest are rated High, Medium, or Low. The engagement strategy follows from where each stakeholder sits on those two axes.

Stakeholder	Role	Infl.	Int.	Needs / expectations	Impact	Engagement
Cold-storage operators	Buyer; run the facilities	High	High	Less spoilage; pass audits; fair price	Positive: adoption, expansion	Reference-site pilot; ROI case; land-and-expand
Duty / quality managers	Daily user of alerts and reports	Med	High	Reliable alerts; simple audit export	Positive: advocacy inside account	Onboarding, training, responsive support
Investors	Provide pre-seed funding	High	High	Recurring revenue; clear path	Positive: capital; negative: withdrawal	Regular updates; ARR and NRR reporting
Sensor suppliers	Supply hardware / gateways	Med	Med	Volume; joint sales	Positive: bundled channel	Multi-supplier contracts; co-selling
Regulators (DM; pharma)	Set audit standards	High	Med	Compliant, accurate records	Positive: legitimacy; negative: rejection	Templates mapped to their requirements
Pharma buyers / 3PLs	Downstream quality gate	High	Med	GDP-grade proof per consignment	Positive: pulls record through chain	Enter after food-side references exist

Stakeholder	Role	Infl.	Int.	Needs / expectations	Impact	Engagement
Cargo insurers	Potential validation partner	Med	Med	Lower claims risk	Positive: distribution channel	Data-sharing pilots; referral terms
Competitors	Global monitoring vendors	Low	High	Defend pharma accounts	Negative: pricing pressure	Benchmark; avoid direct engagement

Table 10. Stakeholder map (DM = Dubai Municipality).

14.4 Innovation Matrix

ColdProof uses largely existing technology — IoT sensors, cloud infrastructure, and a software subscription — but introduces a new business model: a per-point compliance record aimed at the under-served UAE food mid-market that pharmaceutical-grade incumbents overlook. On the matrix this places it in the Disruptive quadrant (new business model serving a low-end or existing market), bordering Architectural because it also reconfigures existing tools for a specific UAE food-compliance niche. It is not Radical: neither the technology nor the market is new in kind. This is the low-end entry pattern — serve the segment incumbents ignore first, then move upmarket into pharmaceuticals — which is why the launch is framed as disruptive. The classification is an analytical judgement, not a sourced fact.

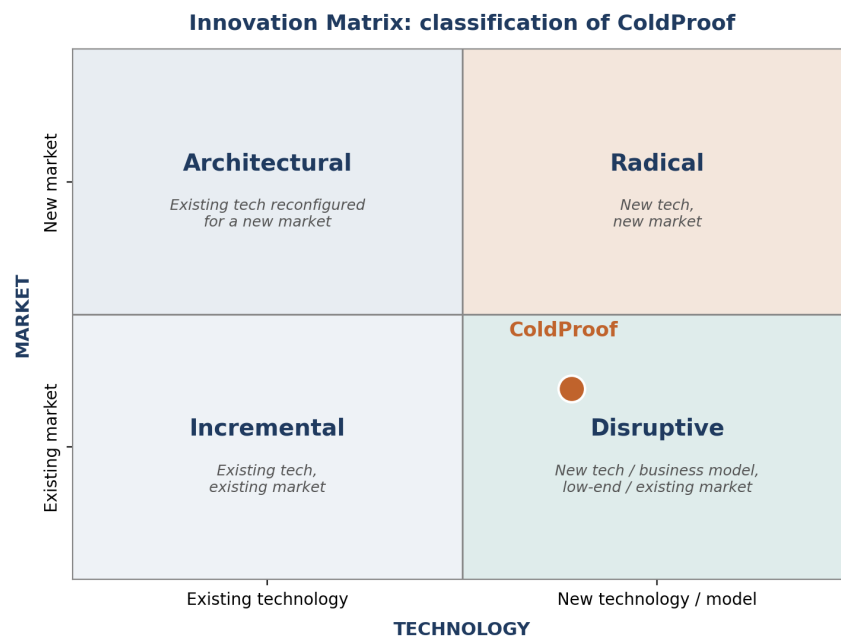


Figure 3. Innovation Matrix classification of ColdProof (X-axis technology, Y-axis market).

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16. Appendices

The Key assumptions register (financial model) is in [Appendix A](#). The figures in sections 9 and 10 are based on the following assumptions, which need to be proven in the first deployments of the venture; an average of 200 monitored points; a price of AED 40 per point per month (mid-point of the AED 30–80 range); an increase in accounts from 6 to 60 over 5 years; gross margin of AED 70–80% after hardware pass-through; as well as a build cost of AED 0.8–1.5 million for the first version. Pharmaceutical add-on is for pharmaceutical lanes only and increases by AED 50-200 per shipment, over the 5 years and is assumed to be a relatively small uplift factor ranging from 1.00 to 1.28.

[Appendix B](#). Divergence in published market size estimates. There are different published estimates depending on the scope. The global cold-chain monitoring market falls between about USD 7–8 billion and an amount that's several times greater if you consider the market to be extended to include not only hardware and services, but also adjacent logistics. According to IMARC Group (2025), the UAE cold-chain logistics market will reach USD 1.4 billion in 2024, whereas Verified Market Research (2025) estimates it to be valued at USD 2.5 billion in 2024, both have the market growing in the 7–8% range. The numbers used in this report are the smaller, more pessimistic estimates for planning purposes and the higher numbers are considered to be conservative upper limits.

[C. Glossary of metrics](#). Active subscribers/Annual recurring revenue (ARR): value of active subscriptions calculated on an annual basis. Monthly recurring revenue (MRR): monthly equivalent. Average revenue per account (ARPA): Recurring revenue / Number of Accounts. Net revenue retention: the revenue saved and increased from already established accounts. CAC payback: (months) Gross margin needed to offset the cost of an account. The point being monitored: one room, chamber or vehicle being monitored